

Video Generation with Predictive Latents

Yian Zhao, Feng Wang, Qiushan Guo, Chang Liu, Xiangyang Ji, Jian Zhang et al. | ByteDance Seed
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Research Questions

- How can video VAEs encode temporally predictive structures to improve diffusability for video generation?
- What is the relationship between predictive accuracy and generative quality in video latent spaces?
- How does predictive reconstruction learning enhance motion-aware representations in video VAEs?

Core Idea

Video VAEs typically optimize only for reconstruction quality, but this doesn't translate to better generative performance. The key insight is that forcing the model to predict future frames from partial observations enriches the latent space with temporal dynamics and motion priors. The method randomly drops future frames during training, requiring the decoder to reconstruct the complete video from partial encoded information, which compels the latent space to encode predictive temporal structures rather than just static visual content.

Key Formula

PV-VAE Loss Function

$$L_{total} = \lambda_{rec}(L_{MSE} + L_{Diff}) + \lambda_{lips}L_{LPIPS} + \lambda_{gan}L_{GAN} + \lambda_{kl}L_{KL}$$

Total loss combining reconstruction (MSE), temporal difference loss (Diff), perceptual loss (LPIPS), adversarial loss (GAN), and KL regularization. The temporal difference term L_{Diff} prevents copy-shortcut by requiring reconstruction of frame differences, forcing the model to learn motion dynamics rather than static content copying.

Key Figure

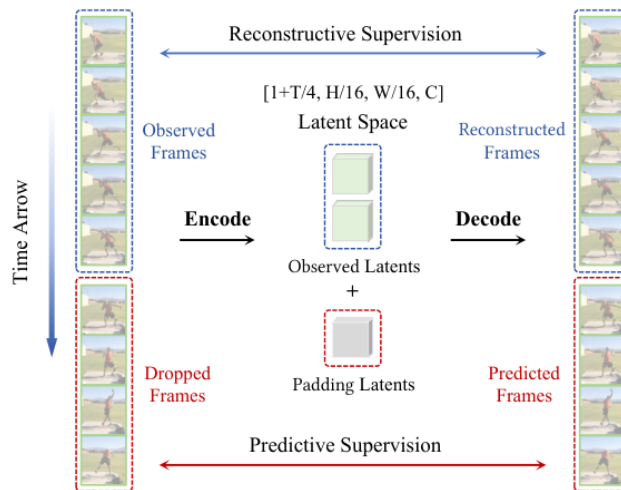


Figure 2 Overall pipeline of the proposed PV-VAE. PV-VAE randomly discards future frames and encodes only observed ones. The padded latents are then decoded to reconstruct the full video, enabling the model to learn visual reconstruction and temporal understanding jointly from reconstructive and predictive supervision.

Figure 2: Shows the complete PV-VAE pipeline where future frames are randomly dropped, only observed frames are encoded, padding latents fill the temporal gaps, and the decoder reconstructs the full video with both reconstructive and predictive supervision.

Limitations

- Exhibits subtle limitations in reconstructing dense text due to scarcity of text-heavy samples in training data
- Requires additional decoder fine-tuning stage to bridge training-inference gap, adding computational overhead
- Motion-aware objective may not generalize well to static or low-motion video content
- Transformer-based variant shows limited generative capability compared to CNN-based implementation despite computational efficiency gains